

E-Commerce Operations



Drop Shipping, Order Tracking, Payment Management
& Payment Gateways

 Drop Shipping  Order Tracking  Payment Systems

Course Overview

01 Introduction

The Modern E-Commerce
Ecosystem

Understanding how drop shipping, order tracking, payment management, and payment gateways interconnect to create seamless customer experiences in the digital marketplace.

🕒 15 minutes

02 Drop Shipping

Fundamentals & Business
Model

Business model mechanics, platform comparisons, supplier networks, pricing strategies, and profit optimization with real-world margin calculations.

🕒 35 minutes

03 Order Tracking

Systems &
Technologies

Real-time visibility solutions, WMS/TMS integration, fulfillment process flow, and implementation strategies with case studies.

🕒 30 minutes

04 Payment Management

Security & Compliance

Payment methods landscape, PCI DSS compliance framework, fraud prevention strategies, and chargeback management best practices.

🕒 25 minutes

05 Payment Gateways

Integration & Best
Practices

The Modern E-Commerce Ecosystem

🔗 Interconnected Operations

Modern e-commerce success depends on the seamless integration of four critical operational pillars. Each component must work in harmony to deliver the speed, transparency, and security that today's consumers demand.

📦 Drop Shipping

Inventory-free fulfillment model

🚚 Order Tracking

Real-time visibility systems

🛡️ Payment Security

PCI DSS compliance & fraud prevention

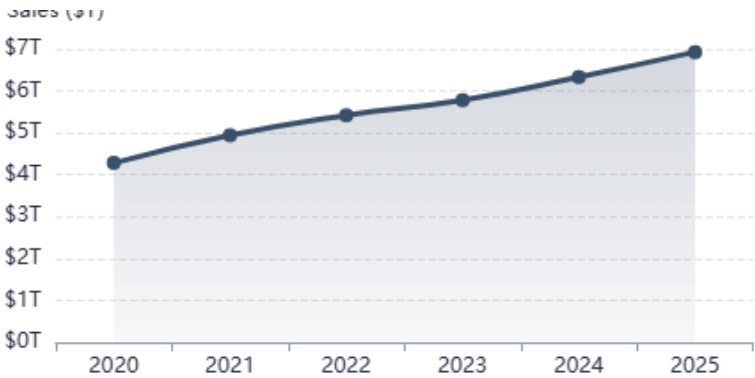
💳 Payment Gateways

Transaction processing infrastructure

💡 The Customer Experience Imperative

A breakdown in any single operational component creates ripple effects across the entire customer journey. **90% of consumers expect order tracking**, and **76% abandon carts** when their preferred payment method isn't available. Operational excellence isn't optional—it's the competitive differentiator.

Global E-Commerce Growth



Source: eMarketer, Statista Global Consumer Survey 2024

Key Market Statistics

\$6.3T

Global E-Commerce Sales
2024

+9.4%

YoY
Growth

21%

of Total Retail
Sales

2024

Market
Share

2.8B

Digital Buyers
Worldwide

-35%

Global
Population

Drop shipment Model (Simplified)

MODULE 01

Drop Shipping Fundamentals

Understanding the Business Model, Processes, and Platforms

4

Core Process Steps

10-40%

Typical Gross Margins

\$362B

Market Size 2025

Retailer

The order is automatically sent to the drop shipping supplier

Third-Party Supplier

The supplier processes and prepares the order

The order is delivered directly to customer

What is Drop Shipping? The Business Model Explained

The Four-Step Process

- 1 Customer Places Order**
A shopper browses your online store, selects a product, and completes checkout. You're the seller of record, but hold no inventory.
- 2 Order Sent to Supplier**
Order details (SKU, quantity, address, payment confirmation) are automatically transmitted to your chosen supplier.
- 3 Supplier Ships Product**
The supplier picks, packs, and ships directly to the customer. They handle all logistics while you maintain brand control.
- 4 You Manage Experience**
You own the customer relationship—marketing, support, order updates, and loyalty campaigns drive repeat purchases.

i Key Insight: The retailer never sees or handles the product. Profit comes from the difference between retail price and supplier cost, minus operational expenses.

Business Model: Pros & Cons

✓ Advantages

- **Low Startup Costs:** No inventory investment required
- **Location Independent:** Operate from anywhere
- **Wide Product Range:** Test products without commitment
- **Scalable:** Easy to expand product catalog
- **Reduced Risk:** No unsold inventory losses

✗ Challenges

- **Lower Margins:** 10–40% vs 50%+ traditional retail
- **Inventory Issues:** Stockouts, quality control gaps
- **Shipping Complexity:** Multiple suppliers = higher costs
- **Supplier Dependence:** Your reputation at their mercy
- **High Competition:** Easy entry = crowded market

Key Players in the Ecosystem



Retailer (You)

Store owner, marketer, customer service provider



Supplier

Manufacturer or wholesaler handling fulfillment



Customer

End buyer expecting seamless experience



Carrier

Shipping partner (USPS, FedEx, UPS, DHL)

Drop Shipping Platforms & Supplier Networks



AliExpress

Global marketplace

- + Millions of products
- + Low product costs
- Long shipping times
- Quality inconsistency

Best for: Beginners testing products



Spocket

US/EU suppliers

- + Faster shipping (2-5 days)
- + Higher quality products
- + Branded invoicing
- Higher product costs

Best for: Premium brands



DSers

AliExpress automation

- + Bulk order processing
- + Supplier optimization
- + Free plan available
- Limited to AliExpress

Best for: High-volume sellers

Platform Comparison Matrix

Platform	Pricing	Shipping	Quality	Integration
AliExpress	Free	• 15-45 days	• Variable	DSers, Oberlo
Spocket	\$39.99/mo	• 2-5 days	• High	Shopify, Woo
Zendrop	\$49/mo	• 3-8 days	• High	Shopify
DSers	Free-\$19.90	• 15-45 days	• Variable	Shopify, Woo
Modalyst	\$35/mo	• 3-7 days	• High	Wix, Shopify

Success Metrics

Average Order Value

Sweet spot for profitability

\$45-

75

Conversion Rate

Industry average for dropshipping

1.5-

3%

Customer Acquisition Cost

Via Facebook/Instagram ads

\$15-

30

Return Rate

Higher than traditional retail

8-

15%

Drop Shipping Pricing Strategy & Profit Optimization

Margin Floor Framework

A margin floor is your **minimum acceptable gross margin**—the guardrail that prevents selling at a loss. Without it, hidden costs (returns, ad fluctuations, fees) silently erode profits.

Calculate True Cost Base:

Product Cost	+ Supplier price
Shipping	+ Delivery to customer
Platform Fees	+ Marketplace/Shopify fees
Payment Processing	+ 2.9% + \$0.30 typical
Ad Spend	+ \$15-30 per acquisition
Total Cost	= True Cost Base

10-20%

Low Margin (Electronics)

20-30%

Standard (General)

30-40%

High (Beauty/Wellness)

40%+

Premium (Custom/POD)

💡 The 3X Rule

Products should be sold for **at least 3 times their cost price** to remain profitable after all expenses. Example: \$5 product → minimum \$15 retail price.

Real-World Profit Calculations

Phone Case (Low Margin Trap)

RISKY

Product Cost	Retail Price
\$2.00	\$5.00
Gross Margin	Ad Cost
60%	\$4.50

Net

-

Profit

\$1.50

⚠️ High-volume, low-margin products often lose money after ads

Yoga Mat (Healthy Margin)

PROFITABLE

Product Cost	Retail Price
\$20.00	\$59.00
Gross Margin	Ad Cost
66%	\$12.00

Net

\$27.0

Profit

0

✅ High-perceived-value products sustain healthy margins

Minimalist Watch (Standard)

BALANCED

Product Cost	Retail Price
\$25.00	\$39.00
Gross Margin	Ad Cost
35%	\$8.00

Net

\$6.0

Profit

0

ℹ️ Requires volume to scale profitably

MODULE 02

Order Tracking Systems

Real-Time Visibility, Technologies,
and Customer Experience

90%

Want Order Tracking

60%

Will Switch Without It

50%

More Likely to Reorder

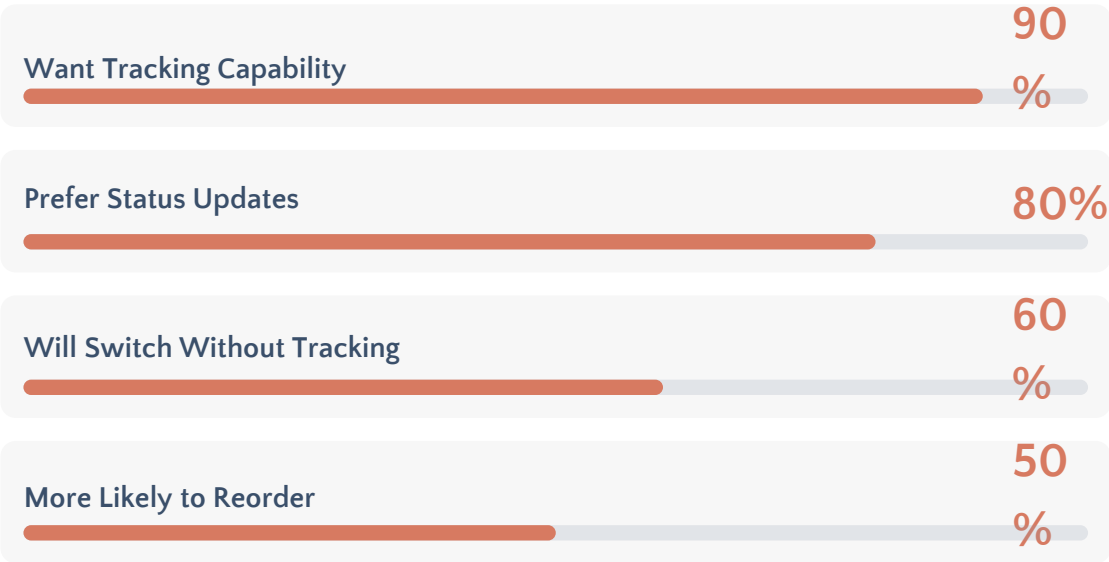
Order Tracking: From Warehouse to Doorstep



Key Technologies & Systems

- **WMS - Warehouse Management System**
Centralizes inventory, order, and shipping information. Connects warehouse to carriers and tracking platforms. Real-time stock synchronization across all channels.
- **TMS - Transportation Management System**
Organizes and optimizes shipments. Maintains real-time communication between company and carriers for accurate tracking.
- **RFID & Barcode Scanning**
Logs each package in real-time, linking tracking numbers to operations. GPS tracking pinpoints shipments during transportation.
- **ERP Integration**
Unites all business processes—sales, procurement, inventory, warehousing, distribution—for end-to-end traceability.

Consumer Expectations



! Critical Insight: Order tracking is no longer a "nice-to-have"—it's essential infrastructure. Businesses without tracking capabilities risk losing the majority of their customers to competitors who offer transparency.

Real-Time Order Tracking Implementation

How Real-Time Tracking Works

- 1

Data Integration

Platform aggregates information from warehouses, carriers, and transportation systems. Includes barcode scans, GPS data, and RFID sensor readings.
- 2

Cloud Synchronization

All data uploaded to secure cloud environment accessible to business and customers. Ensures consistency across web, mobile, and API integrations.
- 3

Predictive Analytics

AI tools analyze trends to predict delivery delays, estimate times, and suggest optimal routes or couriers based on historical performance.
- 4

Customer Updates

Automated notifications via SMS, email, or web dashboards provide complete transparency throughout the delivery cycle.

Business Benefits

20%
Logistics cost reduction

50%
Stockout reduction

30%
Emergency shipping decrease

15-25%
Inventory carrying cost savings

Case Studies: Implementation Success

Wallbox - EV Chargers

Challenge: Managing growing volume of international orders while tracking every shipment in real-time.

Solution: Implemented Easy WMS with Multi-Carrier Shipping Software module, integrating DHL, UPS, GLS.

- Results:**
- Real-time visibility from fulfillment to last-mile
 - Streamlined international distribution
 - Reduced logistics costs through data-driven planning

Stand-Privé.com - Fashion E-commerce

Challenge: Processing thousands of daily orders with full traceability, especially during peak season (Black Friday).

Solution: Deployed Easy WMS integrated with ERP and e-commerce platforms for automated order processing.

- Results:**
- Reduced shipping times significantly
 - Detailed tracking information for customers
 - Proactive issue management during peak sales
 - Strengthened customer loyalty

Implementation Best Practices

- ✓

Choose scalable logistics partner
- ✓

Train staff on visibility data
- ✓

Integrate systems via APIs
- ✓

Monitor performance metrics

MODULE 03

Payment Management

Secure Transactions, Compliance,
and Fraud Prevention

\$362B

Fraud Cost 2023-2028

86%

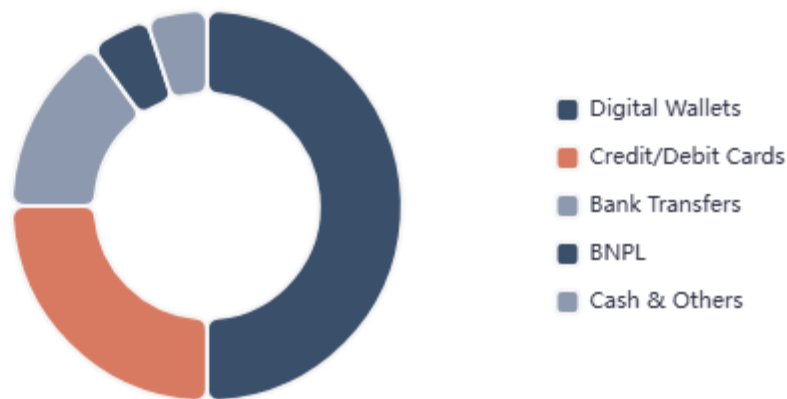
Friendly Fraud Rate

45%

Chargeback Win Rate

Payment Methods Landscape 2024-2025

Global Payment Method Market Share



Source: Worldline Payment Trends 2025, McKinsey Digital Payments Survey

Digital Wallet Dominance

Digital wallets have overtaken cards globally due to convenience and mobile integration. 80% of Gen Z and Millennials prioritize mobile payment options.

50%+
Use digital wallets regularly

76%
Abandon without preferred method

Payment Method Breakdown



Digital Wallets

Apple Pay, Google Pay, PayPal, Samsung Pay

-50%



Credit & Debit Cards

Visa, Mastercard, Amex, Discover

-25%



Bank Transfers

ACH, UPI (India), FedNow (USA), SEPA (EU)

-15%



Buy Now, Pay Later

Klarna, Afterpay, Affirm, Sezzle

-5%



Cash & Others

Cash on delivery, cryptocurrency, checks

-5%

BNPL Milestone

In Q4 2024, BNPL usage among consumers under 40 exceeded credit card usage for the first time. By H1 2025, 50% of younger consumers used BNPL in the previous 3 months—up 6 points from a year ago.

PCI DSS Compliance & Payment Security

PCI DSS 4.0 Framework

The Payment Card Industry Data Security Standard (PCI DSS) ensures security of cardholder data throughout its lifecycle. E-commerce platforms must protect sensitive information from transaction to data processing end.

🔒 Data Encryption

Encrypt data during transmission (TLS 1.2+) and storage to prevent unauthorized access

👤 Access Controls

Limit access to cardholder data to only individuals whose job requires such access

🔍 Monitoring & Testing

Regularly test security systems and processes to ensure effectiveness

PCI DSS 4.0: New Client-Side Focus

Requirement 11 introduces new client-side protections emphasizing code integrity and tamper detection for payment scripts.

Code Integrity

Ensure integrity of browser-running scripts

Tamper Detection

Alert on unauthorized script modifications

Three-Tier Authorization Model

TIER 1 Critical Infrastructure

Scripts touching checkout functionality or cardholder data. Highest scrutiny required.

- ✓ VP or CISO approval required
- ✓ SRI (Subresource Integrity) mandatory
- ✓ Strict change control processes

TIER 2 Revenue-Critical Features

Analytics, conversion tracking, fraud detection. Important for growth but no direct payment data handling.

- ✓ Monthly review schedule
- ✓ Logged within CSP
- ✓ Continuous monitoring

TIER 3 Non-Critical Scripts

General website functionality, non-revenue features. Standard security practices apply.

- ✓ Quarterly review
- ✓ Basic monitoring

Compliance Scope Levels

SAQ A

Lowest scope
Hosted pages/iFrames

SAQ A-EP

Medium scope
Tokenization

SAQ D

Highest scope
Direct API integration

Fraud Prevention & Chargeback Management

E-Commerce Fraud Landscape

\$362B

Projected Cost 2023-2028

\$48B

Damages in 2023 Alone

86%

Friendly Fraud Rate

60%

Say Fraud is Increasing

Common Fraud Warning Signs

- ⚠️ Unusually large purchases or multiple orders from same IP
- ⚠️ Mismatched billing and shipping details
- ⚠️ Transactions at odd hours, repeated card declines before success
- ⚠️ Rapid account creation with disposable emails

Prevention Strategies



3D Secure Authentication

Extra security layer requiring customer authentication via text or biometric



Real-Time Monitoring

AI-powered systems analyze purchasing patterns and flag unusual behavior



AVS & Geolocation

Address verification and IP tracking to identify suspicious locations



Transaction Limits

Spending limits and additional checks for high-value purchases

Chargeback Management

Critical Reality: Merchants win only **45% of chargeback disputes**. Prevention is significantly more cost-effective than disputing after the fact.

Response Framework

- 1 Evaluate the Dispute**
Review chargeback details and reason code to determine validity
- 2 Decide Whether to Dispute**
Compare transaction value, fees, and available evidence
- 3 Prepare Evidence**
Collect documentation supporting the transaction and reason code
- 4 Submit On Time**
Track deadlines—missed deadlines result in automatic losses
- 5 Review Outcomes**
Monitor results and identify recurring issues for prevention

Key Metrics to Track

Chargeback Ratio
vs total transactions

Win Rate
Successfully disputed

Fraud vs Non-Fraud
Dispute breakdown

Avg Dispute Cost
Including fees

Payment Gateways

Integration, Comparison,
and Best Practices

2.9%

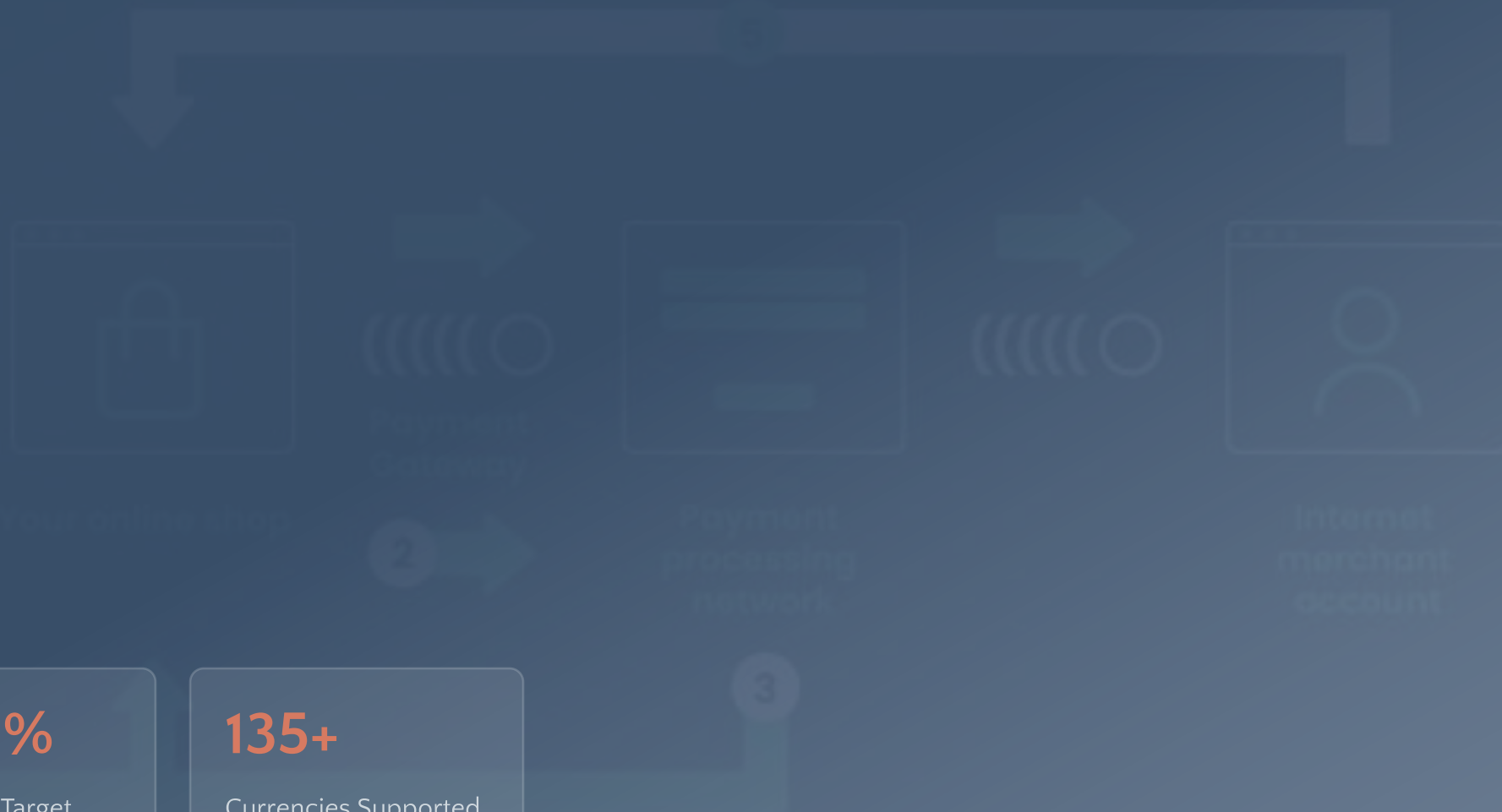
Typical Processing Fee

99.99%

Uptime SLA Target

135+

Currencies Supported



Major Payment Gateways: Stripe vs PayPal vs Square

S

Stripe

Developer-first

2.9% + \$0.30

Online card processing

+ Best API & documentation

+ 135+ currencies

+ Advanced subscriptions

- Steep learning curve

Best for: SaaS, subscriptions, global businesses

P

PayPal

Consumer trust leader

2.99% + \$0.49

Standard processing

+ 400M+ active accounts

+ High conversion rates

+ Instant brand recognition

- Higher fees than competitors

Best for: Consumer trust, conversion optimization

Sq

Square

Omnichannel solution

3.3% + \$0.30

Online (free tier)

+ Best POS integration

+ Free plan available

+ All-in-one ecosystem

- Higher online rates

Best for: Retail, restaurants, omnichannel

Cost Comparison by Order Value

Order Value	Stripe	Square	PayPal	Cheapest
\$20	\$0.88	\$0.96	\$1.09	Stripe
\$50	\$1.75	\$1.95	\$1.99	Stripe
\$100	\$3.20	\$3.60	\$3.48	Stripe
\$500	\$14.80	\$16.80	\$15.44	PayPal

i

At lower tickets, fixed cents matter. At higher tickets, percentage dominates.

💡 Selection Guide 2025

Mostly In-Person (Retail, Cafe)

Use Square as primary. Cohesive POS, competitive in-person rates, instant deposits.

SaaS/Subscriptions/Developers

Use Stripe as backbone. Mature billing tools, transparent pricing, easy global expansion.

Consumer E-Commerce

Use Stripe/Square for cards, but keep PayPal at checkout for trust and conversion.

Mixed Physical + Online

Square in-store + Stripe online + PayPal optional. Uses each where they shine.

Payment Gateway API Integration

Payment API Types

- **Transaction APIs**
Handle one-time payments, authorizations, captures, and refunds. Used in most e-commerce checkouts.
- **Subscription APIs**
Designed for recurring billing. Manage plans, billing cycles, free trials, and automated retries for failed charges.
- **Reporting & Reconciliation APIs**
Provide transaction data for accounting systems, analytics dashboards, and financial reporting.
- **Payout APIs**
Used by marketplaces and platforms to send money to vendors, freelancers, or partners.

- **Security Prerequisites**
- ✓

Active **SSL Certificate**: Mandatory for encrypting data in transit.
- ✓

TLS 1.2+: Enforce minimum TLS version for all communications.
- ✓

Tokenization: Never store raw card data on your servers.
- ✓

PCI DSS Compliance: Scope depends on integration method.

Integration Roadmap

- 1

Obtain API Credentials
Create developer account, access test and live API keys, review documentation
- 2

Set Up Sandbox Environment
Configure local environment, use test card numbers to simulate approvals/declines
- 3

Implement Tokenization
Frontend sends card details directly to gateway, backend uses secure token to charge
- 4

Handle Webhooks
Create webhook endpoints for async updates, use idempotency keys, verify signatures
- 5

Production Launch
Switch to live credentials, monitor transactions, implement error handling

Critical Features to Evaluate

- 99.99% Uptime**
Reliability SLA

Strong SDKs
Developer experience
- Multi-Currency**
Global support

Fraud Detection
Built-in security
- 3D Secure**
Authentication

Detailed Reporting
Reconciliation

 **Pro Tip:** Treat webhooks as the final source of truth for transaction state. Implement proper error handling and retry mechanisms for failed events.

Key Takeaways

1 Drop Shipping Success

Requires strong supplier relationships, margin discipline (3X rule), and quality over quantity. Success comes from selling fewer SKUs with stronger brands, not racing to the bottom on price.

2 Order Tracking is Essential

90% of consumers demand tracking, and 60% will switch to competitors without it. Real-time visibility through WMS/TMS integration is now infrastructure, not a feature.

3 Payment Security is Non-Negotiable

PCI DSS compliance protects your business and customers. Implement three-tier authorization, tokenization, and fraud prevention. Security breaches cost more than compliance.

4 Gateway Selection Matters

Choose based on business model: Stripe for SaaS/subscriptions, Square for retail/omnichannel, PayPal for consumer trust. Consider mixed models for optimal coverage.

📋 Immediate Action Steps

- ✓ Audit current payment security and PCI compliance status
- ✓ Evaluate order tracking capabilities against customer expectations
- ✓ Review payment gateway costs and optimize for your transaction mix